

SoccerNet-MVFoul — Dataset Reference

Prepared for the **Hakam** capstone project. Everything below is needed before preprocessing begins. Sources: the official SoccerNet/sn-mvfoul repository, its `VARS model/config/classes.py`, and the sample `annotations.json` shipped with the interface.

1. Overview

Property	Value
Content	Multi-view foul incidents from professional football matches
Total actions	3,901 from 500 matches
Leagues / seasons	6 European leagues, 2014–2017
Views per action	At least two — one live action plus one or more replays
Annotator	A professional referee with 300+ official matches
Properties per action	10 referee-perspective attributes
Licence	NDA-gated. Research and non-commercial use. Redistribution prohibited.

The labels are a qualified referee's judgement, which is what makes this dataset unusual: the model reproduces documented expert decisions rather than an objective measurement. It is also why perfect accuracy is unreachable — referees disagree on borderline challenges, and that disagreement is baked into the ground truth.

2. Splits

Split	Actions	Folder	Labels
Train	2,916	Train	Yes
Valid	411	Valid	Yes
Test	301	Test	Yes
Challenge	273	Chall	No — held out for the benchmark

Use the official splits exactly as they ship. Do not re-split. Preserve the folder names — the loader depends on them. The test set is only 301 actions, so reported metrics carry real variance; this belongs in the error analysis.

3. File structure

`annotations.json`:

```
{ "Set": "train", "Number of actions": N,  
  "Actions": { "0": {...}, "1": {...} } }
```

`Dataset/action_<id>/clip_<n>.mp4`

4. Action-level fields

Field	Permitted values	Note
UrlLocal	league \ season \ match	Provenance. Useful for group-aware splitting.
Offence	Offence · No offence · Between	Three values, not two
Contact	With contact · Without contact	
Bodypart	Upper body · Under body	
Upper body part	Use of arms · Use of shoulder · <i>empty</i>	Only when Bodypart = Upper body

Field	Permitted values	Note
Action class	Tackling · Standing tackling · High leg · Holding · Pushing · Elbowing · Challenge · Dive · Dont know	9 classes, incl. an explicit unknown
Severity	1.0 – 5.0	See section 5
Multiple fouls	Yes · No	
Try to play	Yes · No	Central to careless vs reckless
Touch ball	Yes · No	
Handball	Handball · No handball	
Handball offence	<i>often empty</i>	
Clips[]	list of clip objects	See section 6

5. Severity — the field that shapes the model design

Raw	Meaning	Definite?
1.0	No card	Yes
2.0	Borderline No / Yellow	No
3.0	Yellow card	Yes
4.0	Borderline Yellow / Red	No
5.0	Red card	Yes

Two of the five severity values are explicitly borderline, and the Offence field carries a matching third value, *Between*. These encode the annotator's own uncertainty as a label. A decision is required before preprocessing:

Option	Effect
Drop ambiguous labels from training	Cleanest signal. Smaller dataset. Discards exactly the hard cases the product exists to explain.
Merge into the nearest definite class	Keeps all data, injects label noise into the classes that matter most.
Keep as their own class	Most honest, hardest to learn, lowers headline numbers.

Recommended: drop *Between* and severities 2.0 / 4.0 from training, then report performance on them separately as a limitations section. This keeps the cascade stages clean and supplies the evidence for why state of the art on this dataset sits near 50%.

6. Clip-level fields

Field	Example	Note
Url	Dataset/action_0/clip_0	Relative path to the .mp4
Camera type	Main camera center · Close-up player or field referee	Live vs replay angle
Timestamp	1259839	Position within the match
Replay speed	1.0 · 3.6 · 4.2	1.0 = live. Slow-motion replays have higher values.

Replay speed matters. Views of the same incident run at different speeds, so apparent motion differs across views of a single action. Any model reasoning about impact velocity must account for this rather than treating all views as equivalent.

7. Benchmark task definitions

Task	Classes	Published baseline
offence_severity_class	No offence · Offence+No card · Offence+Yellow · Offence+Red	34.5% balanced accuracy
action_class	9 foul types	39.5% balanced accuracy

VARS, published at a CVPR 2023 workshop, reaches roughly 50% on foul type and 46% on the correct sanction. Classes are heavily imbalanced — red cards are rare — which is why the benchmark reports **balanced accuracy** rather than plain accuracy. Plain accuracy would look considerably better and mean considerably less.

8. Mapping to the Hakam cascade

Stage	Question	Derived from	Chance floor
1	Is it an offence?	Offence field (excluding <i>Between</i>)	50%
2	Does it warrant a card?	Severity 1.0 vs 3.0/5.0	50%
3	Yellow or red?	Severity 3.0 vs 5.0	50%

Three binary decisions rather than one multi-class head. Each stage has a 50% chance floor instead of 12.5%, which is what makes roughly 70% per stage attainable where a single 9-way head reports near 40% for the same underlying model quality. Always name the stage when quoting a number — a bare “70% accuracy” invites the assumption that it refers to the full multi-class task.

The remaining fields — Action class, Bodypart, Contact, Try to play, Touch ball — are trained as auxiliary heads. On 3,901 samples this acts as free regularisation, and the predicted attributes give the language model concrete detail to ground its explanation in.

9. Access

1. Sign the NDA at the SoccerNet request form.
2. A password arrives by email after approval.
3. `pip install SoccerNet`

```
from SoccerNet.Downloader import SoccerNetDownloader as Sndl
mySndl = Sndl(LocalDirectory="hakam/data")
mySndl.downloadDataTask(task="mvfouls",
    split=["train", "valid", "test", "challenge"],
    password="...")
```

The task string is **mvfouls**, plural — the repository name drops the s. Standard definition is the default; pass version="720p" for higher resolution. The dataset must never be committed to the repository: the NDA prohibits redistribution.